

# PRATIK NANASO RAUT

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## PROFESSIONAL SUMMARY

MCA graduate with strong proficiency in C/C++, object-oriented design, data structures, and Linux system programming. Built multi-threaded network servers using POSIX threads and TCP sockets, virtual file systems with inode-based memory management, and a custom SQL engine using C++ STL. Experienced in Linux/Unix development with Git version control, Agile/Scrum workflows, and build automation. Solid understanding of memory management, process internals, and network protocols (TCP/IP, UDP).

## TECHNICAL SKILLS

**C/C++ & Systems:** C, C++ (OOP, STL, Templates), Data Structures, Algorithms, Software Design Patterns

**Multi-Threading & Networking:** POSIX Threads, Mutex, Semaphores, TCP/IP Socket Programming, UDP, Concurrency

**Memory & OS Internals:** Memory Management, /proc Filesystem, Virtual Address Space, Inode Management, File Locking

**Linux & DevOps:** Linux (Ubuntu), Bash Scripting, POSIX System Calls, Shell Automation, Git, Maven, CI/CD Concepts

**Backend & Databases:** Java, Spring Boot, REST APIs, MySQL, PostgreSQL, MongoDB, SQL Query Optimization

**Tools & Methodologies:** Git, GitHub, Postman, VS Code, IntelliJ IDEA, Agile/Scrum, Code Reviews, Unit Testing

## PROFESSIONAL EXPERIENCE

### SetTribe IT Solutions

Feb 2026 – Jul 2026

*Software Developer Intern*

*Pune, Maharashtra*

- Developed application modules using Java, Spring Boot, and MySQL within Agile/Scrum sprint cycles with daily standups.
- Designed and tested RESTful API endpoints with JWT authentication, input validation, and role-based access controls.
- Managed Git-based version control workflows including branching, merging, code reviews, and Maven build pipelines.
- Operated on Linux development environments using Bash scripting, CLI tools, and automated build/test routines.

## PROJECTS

### Concurrent FTP Server – Multi-Client File Transfer | C, POSIX Threads, TCP/IP Sockets, Multithreading

- Engineered concurrent client handling using POSIX threads over TCP sockets, supporting simultaneous file transfers without data corruption.
- Implemented thread-safe resource management using mutex and semaphore primitives, eliminating race conditions under concurrent load.
- Designed application-layer protocol over raw sockets handling authentication, file listing, upload, and download command sequences.

### Customised Virtual File System (CVFS) | C++, OOP, Data Structures, OS Concepts, Memory Management

- Simulated Linux file system operations (open, read, write, lseek) in C++ using inode-based metadata management and a custom interactive shell.
- Achieved O(1) file lookup by designing hash table-based inode indexing with chaining for collision resolution.
- Implemented in-memory file allocation table using C++ classes managing file permissions, ownership flags, and directory traversal.

### Customized DBMS – Relational SQL Engine | C++, OOP, STL Containers, Data Structures, SQL

- Designed SQL query parser in C++ using OOP class hierarchy, supporting full DDL (CREATE, DROP) and DML (INSERT, SELECT, UPDATE, DELETE).
- Engineered B-tree indexing structure for accelerated record lookup, reducing sequential scan overhead on large in-memory datasets.
- Simulated multi-table relational schema with join operations, constraint enforcement, and memory-managed record storage using C++ STL.

### ProcMemInspector – Linux Process Memory Analyzer | C, /proc Filesystem, Linux Internals, Memory Management

- Developed /proc filesystem parser in C to extract virtual memory layout of live Linux processes (stack, heap, BSS, data, shared libraries).
- Implemented process statistics monitoring by reading /proc/[pid]/maps and /proc/[pid]/status with structured output formatting.
- Analyzed virtual address space segmentation to demonstrate memory isolation boundaries and shared library loading behavior.

### Command Suite – Linux Core Utilities Re-implementation | C, POSIX System Calls, Linux System Programming

- Re-implemented core Linux utilities (ls, ps, mv, touch, uname) using POSIX system calls including stat(), opendir(), fork(), getpid().
- Handled file permission bitmask parsing, directory traversal, process status extraction, and GNU-style argument parsing.

## EDUCATION

### Dnyansagar Institute of Management and Research, Pune (SPPU)

2024 – 2026

*Master of Computer Applications (MCA) – CGPA: 7.32*

*Pune, Maharashtra*

### C.T. Bora College, Shirur (SPPU)

2021 – 2024

*Bachelor of Computer Applications (BCA) – Marks: 73.34%*

*Pune, Maharashtra*